

Blériot XI Monoplane



“No pilot of today, no matter how great, could repeat this exploit [the cross-Channel flight] in such an aircraft and with such an engine.”

CHARLES DOLLFUSS

AVIATION HISTORIAN, WRITING IN 1932

THE BLÉRIOT XI MONOPLANE in which Louis Blériot made his historic crossing of the English Channel on July 25, 1909, established the main monoplane design features for generations to come, and paved the way for the Blériot company's considerable commercial success.

It had a front-mounted modified motorcycle engine, tricycle undercarriage, front-mounted wings, and rear-mounted tailplane, elevators, and rudder. Its fuselage was a simple wire-braced, wooden-box girder, enclosed with fabric at the front to give the pilot some protection. Strips of rubber bungee were stretched down the front undercarriage legs to absorb landing shocks, while the tailwheel had a stiff spiral spring to perform the same function. Facilitated by wires that ran above and below the fuselage, lateral control was provided by warping the wing's trailing edges, directional control was provided by the moving rudder, and longitudinal control was provided by the moving elevators on the tips of the tailplane. Like all early airplanes, the Blériot XI had thin wings. They were braced against

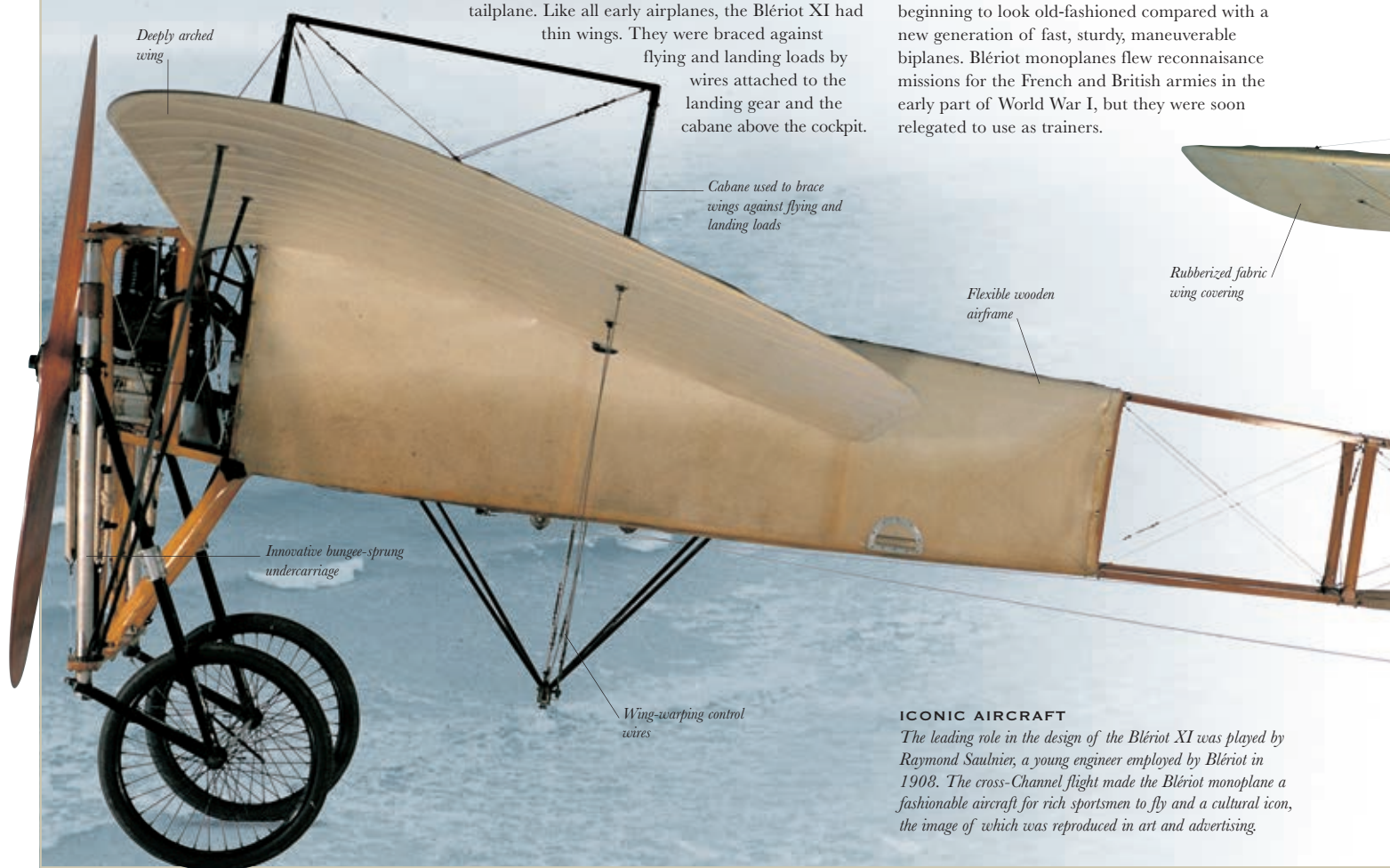
flying and landing loads by wires attached to the landing gear and the cabane above the cockpit.



DESIGN DISADVANTAGE

One of the main disadvantages of all monoplanes, including the Blériot model above, was the weakness of the single wing, which needed strong wire bracing to withstand the loads placed upon it.

After the cross-Channel flight, rich sportsmen lined up to buy Blériot monoplanes. So did armies when they began to investigate the military use of aircraft in 1910–11. Many improvements were made to later versions of the monoplane, including the early replacement of the underpowered Anzani engine with a 50hp Gnome rotary engine. More than 130 Blériot XIs were built, but by 1914 they were beginning to look old-fashioned compared with a new generation of fast, sturdy, maneuverable biplanes. Blériot monoplanes flew reconnaissance missions for the French and British armies in the early part of World War I, but they were soon relegated to use as trainers.



Deeply arched wing

Cabane used to brace wings against flying and landing loads

Flexible wooden airframe

Rubberized fabric wing covering

Innovative bungee-sprung undercarriage

Wing-warping control wires

ICONIC AIRCRAFT

The leading role in the design of the Blériot XI was played by Raymond Saulnier, a young engineer employed by Blériot in 1908. The cross-Channel flight made the Blériot monoplane a fashionable aircraft for rich sportsmen to fly and a cultural icon, the image of which was reproduced in art and advertising.